17										
18	**X**	0	1	2	3	4	5	6	7	8
19	**f**	20	100	190	280	350	300	220	120	3
20										
21	**X**	**f**	**expected frequency f(x)=Np(x)**	**Formula**	**Rounded expected frequency**	**Formula**				
22	0	20	5.642978112	=$B$34*BINOMDIST(A22,$B$35,$B$37,0)	6	=ROUND(C22,0)				
23	1	100	46.1822771		46					
24	2	190	165.3561596		165					
25	3	280	338.319759		338					
26	4	350	432.627743		433					
27	5	300	354.0636506		354					
28	6	220	181.1041229		181					
29	7	120	52.93431324		53					
30	8	3	6.768996445		7					
31	Total	1583			1583	=SUM(E22:E30)				
32										

32			
33			Formula
34	N	1583	=B31
35	n	8	
36	np	4.04548326	=SUMPRODUCT(B22:B30,A22:A30)/B31
37	p	0.505685407	=B36/B35
38	q	0.494314593	=1-B37
39	P(X<3)	0.137196093	=BINOMDIST(2,B35,B37,1)
40	P(X<=3)	0.350916724	=BINOMDIST(3,B35,B37,1)
41	P(X>3)	0.649083276	=1-B40
42	P(X>4)	0.375787166	=1-BINOMDIST(4,B35,B37,1)
43	P(4<X<7)	0.611367983	=BINOMDIST(6,B35,B37,1)-BINOMDIST(3,B35,B37,1)
44			

17				
18	**X**	**P(X)**	**XP(X=x)**	**x^2 p(X=x)**
19			=A20*B20	=A20^2*B20
20	-2	0.2	-0.4	0.8
21	-1	0.1	-0.1	0.1
22	1	0.4	0.4	0.4
23	1	0.2	0.2	0.2
24	2	0.1	0.2	0.4
25	Total	1	0.3	1.9
26		=SUM(B20:B2	=SUM(C20:C24)	=SUM(D20:D24)
27				
28		**Value**	**Formula**	
29	**E(X)**	0.3	=SUMPRODUCT(A20:A24,B20:B24)	
30	**E(4X+5)**	6.2	=4*B29+5	
31	**E(X^2)**	1.9	=SUMPRODUCT((A20:A24)^2,B20:B24)	
32	**V(X)**	1.81	=B31-(B29)^2	
33	**V(4X+5)**	28.96	=4^2*B32	
34	**σX**	1.345362405	=SQRT(B32)	
35	**σ(4X+5)**	5.381449619	=SQRT(B33)	
36				

17							
18	**project**	**profit(rs '000)**	**probability**	**loss(rs '000)**	**probability**	**profit**	**FORMULA**
19	1	35	0.5	15	0.5	10	=PRODUCT(B19:C19)-PRODUCT(D19:E19)
20	2	48	0.3	25	0.7	-3.1	=PRODUCT(B20:C20)-PRODUCT(D20:E20)
21	3	45	0.6	20	0.4	19	=PRODUCT(B21:C21)-PRODUCT(D21:E21)
22							
23							
24	"The maximum expected profit is 19 (rs. `000), which is obtained from project 3. hence we select project 3."						

17				
18	**AGE**	**BACHELOR'S DEGREE (B)**	**MASTER'S DEGREE(M)**	**TOTAL**
19	UNDER 30 (A)	40	10	50
20	30-40 (W)	20	30	50
21	OVER 40 ©	90	10	100
22	TOTAL	150	50	200
23		=SUM(B19:B21)	=SUM(C19:C21)	=SUM(D19:D21)
24				
25				
26			**values**	**Formula**
27	**a.**	**P(B)**	0.75	=B22/$D$22
28	**b.**	**P(MnC)**	0.05	=C21/D22
29	**c.**	**P(A/B)**	0.266666667	=B19/B22
30	**d.**	**P(AnB)**	0.266666667	=B19/B22
31				

17							
18	**GIVEN**	**VALUE**		**VALUE**	**FORMULA**		**VALUE**
19	A	1000	P(M1)	0.166666667	=B19/$B$22	P(D|A)	0.01
20	B	2000	P(M2)	0.333333333	=B20/$B$22	P(D|B)	0.02
21	C	3000	P(M3)	0.5	=B21/$B$22	P(D|C)	0.03
22	TOTAL	6000					
23		=SUM(B19:B21)					
24		**VALUE**	**FORMULA**				
25	P(DnA)	0.001666667	=D19*G19				
26	P(DnB)	0.006666667	=D20*G20				
27	P(DnC)	0.015	=D21*G21				
28	P(A)	0.023333333	=SUM(B25:B27)				
29							
30		**VALUE**	**FORMULA**				
31	P(A|D)	0.001666667	=D19*G19				
32	P(B|D)	0.285714286	=B26/$B$28				
33	P(C|D)	0.642857143	=B27/$B$28				
34							